Page: Date: / /

Fall 2019-2020 ↓

⇉ Q1: Solve the following diff eq: ↓

↳ (a) $y\frac{dy}{dx} = \frac{1}{x}\left(\ln\frac{x}{y} + \ln y\right)$

$y\frac{dy}{dx} = \frac{1}{x}\left(\ln x - \ln y + \ln y\right).$

$y\frac{dy}{dx} = \frac{\ln x}{x} \Rightarrow \int y\,dy = \int \frac{\ln x}{x}\,dx.$

$y^2 = \ln x^2 \rightarrow \boxed{y = \ln x + C}.$ #

↳ (b) $x\frac{dy}{dx} - y = xy^7$ $\left[\div x\right]$

$\frac{dy}{dx} - \frac{1}{x}y = y^7 \rightarrow n = 7$

$\int \frac{1}{x}\cdot(-6)\,dx = -6\ln x = \ln x^{-6}$

$e^{\ln x^{-6}} = x^{-6}$

$y = \frac{1}{x^6}\left[\int x^{-6}\cdot(-6)\,dx + C\right] = x^6\left[\int x^{-6}\cdot(-6)\,dx + C\right].$

$= x^6\left[\frac{-6x^{-5}}{-5} + C\right] = x^6\left[6x^{-5} + C\right].$ #

↳ (c) $\underbrace{(3x^2 + y\cos x)}_{M}dx + \underbrace{(\sin x - 4y^3)}_{N}dy = 0$

$\frac{\partial M}{\partial y} = \cos x \equiv \frac{\partial N}{\partial x} = \cos x$ ∴ Exact Eq.

$\int M\,dx + \int N\,dy = x^3 + y\sin x + y\sin x - y^4 = C$

$\therefore x^3 + y\sin x - y^4 = C.$ #

↳ (d) $\frac{dy}{dx} = \frac{x-y+1}{x-y-1}$ $z = x - y$ $\frac{dz}{dx} = 1 - \frac{dy}{dx}$

$1 - \frac{dz}{dx} = \frac{z+1}{z-1} \Rightarrow \frac{-dz}{dx} = \frac{z+1}{z-1} \times \frac{-1}{-1}$

$\frac{-dz}{dx} = \frac{z+1-z+1}{z-1} = \frac{2}{z-1} \Rightarrow \int -(z-1)\,dz = \int 2\,dx$

$\Rightarrow -\left(\frac{1}{2}z^2 - z\right) = 2x + C.$ #

Page: Date: / /

Q2: Find the General Solution →

↳ (a) $\frac{d^3y}{dx^3} - 3\frac{d^2y}{dx^2} + 6\frac{dy}{dx} - 4y = 0$

$m^3 - 3m^2 + 6m - 4 = 0$

$m(m^2 - 3m + 6 - 4) = 0$

$m(m-2)(m-1) = 0$

$m_1 = 0 \quad m_2 = 2 \quad m_3 = 1$

$\# \{ y = C_1 e^{0x} + C_2 e^{2x} + C_3 e^{x} \} \#$

↳ (b) $\frac{d^2y}{dx^2} + y = \underbrace{\sec(x)}_{\frac{1}{\cos}} \underbrace{\csc(x)}_{\frac{1}{\sin}}$

$y'' + y = 0$

$m^2 + 1 = 0$

$m_{1,2} = \pm i$

$y_{c.f} = e^{0x}[C_1 \cos x + C_2 \sin x]$

$u = \cos x \quad v = \sin x$

$u' = -\sin x \quad v' = \cos x$

$uv' - vu' = 1$

$Ax(u) = \int \frac{-v f(x)}{uv' - vu'} = \int \frac{-\sin x \cdot \frac{1}{\cos x} \cdot \frac{1}{\sin x}}{1}$

$= \int \frac{-1}{\cos x} = -\int \sec x \, dx = \frac{-1}{\tan x} \cdot \sec x + C$

$= \frac{\cos x}{\sin x} \cdot \frac{1}{\cos x} = \frac{1}{\sin x} = \boxed{\csc x + C.} \#$

$Bx(v) = \int \frac{u f(x)}{uv' - vu'} = \int \frac{\cos x \cdot \frac{1}{\cos x} \cdot \frac{1}{\sin x}}{1}$

$= \int \frac{1}{\sin x} = \int \csc x = \frac{1}{\cot x} \cdot -\csc x$

$= \frac{\sin x}{\cos x} \cdot \frac{1}{\sin x} = \frac{-1}{\cos x} = \{ -\sec x + K. \} \#$

$y_{g.s} = y_{c.f} + y_{p.i} \quad \#$

$= e^{0x}[C_1 \cos x + C_2 \sin x] +$

$\# \csc x + C - \sec x + K.$

$\int \sec \Rightarrow \ln|\sec| + \ln$

$\int \sec \cdot \frac{\sec x + \tan x}{\sec x + \tan x} = \int \frac{\sec^2 + \sec x \tan x}{\sec x + \tan x}$

$\# \ln|\sec + \tan| \#$

الممسوحة ضوئيا بـ CamScanner

Page: Date: / /

↳ © $\frac{d^2y}{dx^2} - \frac{dy}{dx} = 5x + e^{3x}$ $\quad 5(Ax+B) + Ce^{3x}$

$y'' - y = 0$

$m^2 - 1 = 0$

$m_{1,2} = 1$

$y_{c.f} = C_1 e^x + C_2 x e^x$

$y_{P.I} = 5Ax + 5B + Ce^{3x}$

$y' = 5A + 3Ce^{3x}$

$y'' = 9Ce^{3x}$

$\therefore 9Ce^{3x} - 5A + 3Ce^{3x} = 5x + e^{3x}$

$12C = 1 \quad \{\therefore C = \frac{1}{12}\}$

$-5A = 0 \quad \{\therefore A = 0\}$

$y_{P.I} = \frac{1}{12} e^{3x}$

$\{y_{g.s} = y_{c.f} + y_{P.I} = C_1 e^x + C_2 x e^x + \frac{1}{12} e^{3x}\}$

Q3: @ $\frac{d^2y}{dt^2} - 3\frac{dy}{dt} + 2y = 0 \qquad y(0) = 0, \; y'(0) = 1$

$s^2 y(s) - s\cancel{y(0)} - \overset{1}{\cancel{y'(0)}} - 3s\,y(s) - 3\cancel{y(0)} + 2y(s) = 0$

$y(s)[s^2 - 3s + 2] = 1$

$y(s) = \frac{1}{(s-1)(s-2)} \Rightarrow \frac{A}{s-1} + \frac{B}{s-2}$

$\{*A = -1\} \; \{*B = 1\}$

$\{y(t) = -e^t + e^{2t}\}$

الممسوحة ضوئيا بـ CamScanner

Page: Date: / /

↳ (b) Find the LaPlace Transform

$\frac{1}{2}[\sin 3t - (-\sin t)]$

(i) $f(t) = U_2(t)\, e^3 \sin(t-2) + \sin(t)\cos(2t)$

$$f(s) = e^{-2s}\frac{1}{(s-3)^2+1} + \frac{1}{2s}\left[\frac{3}{s^2+9} + \frac{1}{s^2+1}\right].$$

(ii) $f(t) = t^7 e^{2t} + \int_0^t \sinh\tau \, d\tau$

$$f(s) = \frac{6!}{(s-2)^8} + \frac{1}{s}\left[\frac{1}{s^2-1}\right].$$

↳ (c) Find Inverse LaPlace

(i) $F(s) = e^{-3s}\left[\frac{s}{s^2+10s+5}\right]$ →

$$\frac{s}{s^2+10s+25-25+8} = \frac{s}{(s+5)^2-17}$$

$$f(t) = U_3(t)\cosh\sqrt{17}t.$$

(ii) $F(s) = \mathcal{L}\frac{s+1}{s} = \mathcal{L}(s+1) - \mathcal{L}(s)$

$$F'(s) = \frac{1}{s+1} - \frac{1}{s}$$

$$-t\,F(t) = e^{-t} - 1$$

$$F(t) = \frac{e^{-t}-1}{-t}$$

EBN EMAD 21/2/2021

الممسوحة ضوئيا بـ CamScanner